Powertrain Control - High-Level Design

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1. Component Overview

This document defines the architecture of the Powertrain Control software, responsible for engine and transmission coordi

Software Components

Component: EngineControllerComponent
Component: TransmissionManagerComponent
Component: FuelInjectionComponent

2. Interfaces and Ports

Interface: EngineTorqueInterface
Interface: GearShiftInterface
PPort: EngineController_PPort
RPort: TransmissionManager_RPort
EngineControllerComponent provides EngineTorqueInterface via EngineController_PPort, consumed by TransmissionMar

3. Signals

Signal: EngineSpeed_Signal
Signal: RequestedTorque_Signal
Signal: GearPosition_Signal

4. Dependencies

TransmissionManagerComponent depends on EngineControllerComponent for torque availability before initiating a gear s

5. Functional Flow

EngineControllerComponent computes RequestedTorque_Signal from accelerator pedal input and publishes EngineSpeed